

Chapter 3

QBIO 305 FA26

Summary of lecture slides

Probability

- A probability is a numerical quantity that expresses the likelihood of an event
- The probability of an event E is written $\Pr\{E\}$ or $P(E)$
- $\Pr\{E\}$ is always between 0 and 1 (inclusive)

Frequentist Interpretation of Probability

- $\Pr\{E\}$ = number of times even E occurs divided by number of times chance operation is repeated
- This interpretation only makes sense if the chance operation can be repeated many times

Probability rules

- Probability of an event is always between 0 and 1 (inclusive)
- The probability of all possible events is 1
- The probability an event does not happen is one minus the probability that it does happen
 - The complement of E (denoted E^c) is the even that E does not happen,
 $\Pr\{E^c\} = 1 - \Pr\{E\}$
- For any two events E1 and E2, $\Pr\{E1 \text{ or } E2\} = \Pr\{E1\} + \Pr\{E2\} - \Pr\{E1 \text{ and } E2\}$

Disjoint

- Two events E_1 and E_2 are said to be disjoint (or mutually exclusive) if $\Pr\{E_1 \text{ and } E_2\} = 0$
- Disjoint means 'no overlap'

Conditional probability

- The conditional probability of E2 given E1 is:

$$\Pr(E_2|E_1) = \frac{\Pr(E_2 \text{ and } E_1)}{\Pr(E_1)}$$

Multiplication rule

- For any two events E1 and E2,

$$\Pr\{E1 \text{ and } E2\} = \Pr\{E1\} \times \Pr\{E2|E1\}$$

To clarify...

- $\Pr\{E1 \text{ and } E2\} = \Pr\{E2 \text{ and } E1\}$
- $\Pr\{E1 \text{ or } E2\} = \Pr\{E2 \text{ or } E1\}$
- $\Pr\{E1|E2\} \neq \Pr\{E2|E1\}$

Independence

- Events E_1 and E_2 are independent if $\Pr\{E_1 \text{ and } E_2\} = \Pr\{E_1\} \times \Pr\{E_2\}$
- This is equivalent to $\Pr\{E_2|E_1\} = \Pr\{E_2\}$
- Also equivalent to $\Pr\{E_1|E_2\} = \Pr\{E_1\}$
- In words: two events are independent if knowing that one event occurred does not change the probability of the other event occurring

Disjoint vs independent

- There are exercises to review these concepts in the lecture slides

Law of total probability

- For any two events E and F,

$$P(F) = P(E \text{ and } F) + P(E^c \text{ and } F)$$

- Using the multiplication rule,

$$P(F) = P(F|E)P(E) + P(F|E^c)P(E^c)$$

Bayes' Theorem

- For any two events E and F,

$$P(E|F) = P(F|E)P(E)/P(F)$$

- With the law of total probability,

$$P(E|F) = P(F|E)P(E)/(P(F|E)P(E) + P(F|E^c)P(E^c))$$

Random variable

- Variable that takes on numerical values that depend on the outcome of a chance operation
- Discrete random variable
 - The sum of all the histogram heights is 1
- Continuous random variable
 - The integral under the density curve is 1

Mean of a random variable

- Also known as the expected value or expectation,

$$\mu_Y = E(Y) = \sum_i y_i P(Y = y_i)$$

- The mean is about the average of the random variable distribution
- The mean of a random variable is different from the sample mean
 - Different samples have different sample means but the random variable mean will remain the same (e.g. rolling a die)

Variance and standard deviation of a random variable

$$\sigma_Y^2 = \text{Var}(Y) = \sum_i (y_i - \mu_Y)^2 P(Y = y_i)$$

- Standard deviation (σ_Y) is the square root of the variance
- The variance and standard deviation quantify the spread (or dispersion) of the distribution

Mean and Variance of continuous RV

$$\mu = \int xf(x)dx$$

$$\sigma^2 = \int (x - \mu)^2 f(x)dx$$

A few more things...

- Consider two random variables:
 - X has mean μ_X and variance σ_X^2
 - Y has mean μ_Y and variance σ_Y^2
- Mean of $a + X$: $a + \mu_X$
- Mean of bX : $b\mu_X$
- Mean of $X+Y$: $\mu_X + \mu_Y$
- Mean of $X-Y$: $\mu_X - \mu_Y$
- Variance of $a+bX$: $b^2\sigma_X^2$
 - Standard deviation of $a+bX$: $b\sigma_X$
- Variance of $X+Y$ (assuming independence): $\sigma_X^2 + \sigma_Y^2$
- Variance of $X-Y$ (assuming independence): $\sigma_X^2 + \sigma_Y^2$

Sample statistics

- For constants a and b :

	Y	$a+bY$
Mean	$\bar{y}$	$a + b\bar{y}$
Median	m	$a+bm$
Standard deviation	s	bs
IQR	d	bd

Random variables X and Y

- Where
 - X has mean μ_X and variance σ_X^2
 - Y has mean μ_Y and variance σ_Y^2
 - a and b are constants:

	a+bX	X + Y	X - Y
Mean	$a + b\mu_X$	$\mu_X + \mu_Y$	$\mu_X - \mu_Y$
Variance	$b^2\sigma_X^2$	$\sigma_X^2 + \sigma_Y^2$ if X and Y indep.	$\sigma_X^2 + \sigma_Y^2$ if X and Y indep.